



NATIONAL UNIVERSITY OF
COMPUTER & EMERGING SCIENCES

CARPER — THE CAR PROBLEM SOLVER

Automated Vehicle Damage Detection · Repair-Cost Estimation · Smart Parts Marketplace

CARPER TEAM

SUPERVISOR

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TEAM MEMBERS

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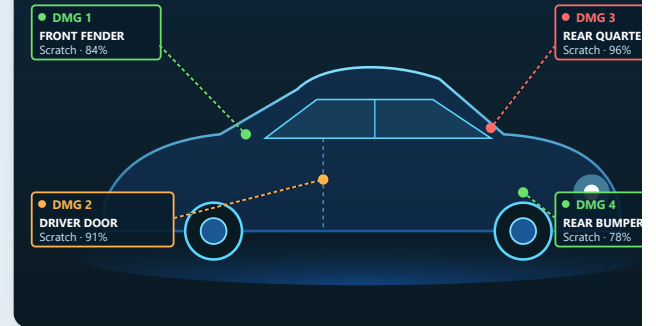
Syed Azhaan Ali (22K-4140)

TEAM LEAD

Syed Wali Mohsin Naqvi (22K-4550)

INSPECTION · 4 issues identified

SCAN



ABSTRACT

Manual car damage assessment is prone to human error, fraud, and is time-consuming — affecting **insurance, vehicle rental, and resale**. **CARPER — The Car Problem Solver** is a smart web-based application that detects exterior vehicle damage and provides an estimated repair cost. It uses the **YOLOv8n** object-detection model trained on the **CarDD** dataset (4,000 high-resolution images, 6 damage classes: dent, scratch, crack, glass shatter, tire flat, lamp broken). For cost estimation, Linear Regression, Random Forest, Extra Trees and Gradient Boosting were evaluated; **Gradient Boosting** was selected. The platform follows a **Client-Server three-tier** architecture: **Next.js · NestJS · PostgreSQL**. Semantic search and price-matching algorithms connect users to affordable replacement parts from online marketplaces.

0.732

MAP@0.5

4.4ms

INFERENCE

4 K

TRAIN IMGS

6

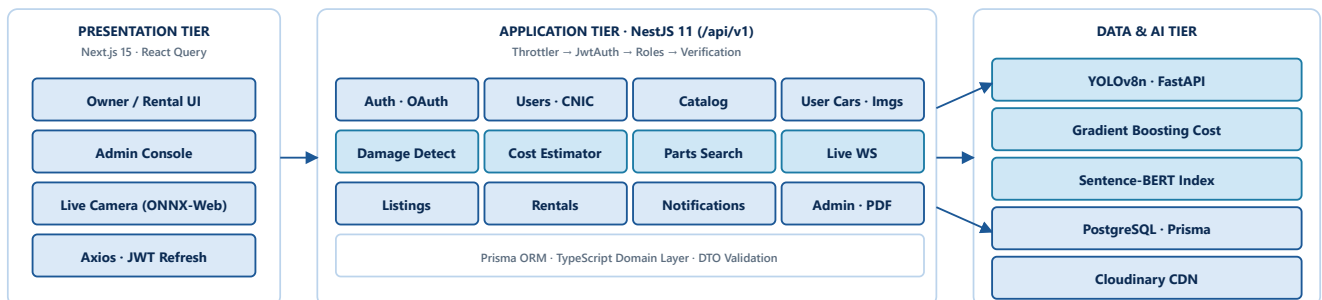
CLASSES

OBJECTIVE

The project delivers a web-based automated vehicle damage inspection

- Detecting **six** external damage types using YOLOv8n on CarDD.
- Estimating repair cost via **Gradient Boosting** regression.
- Recommending affordable parts via **semantic marketplace** search.
- Providing a seamless web interface for owners, rentals & admins.

SYSTEM ARCHITECTURE



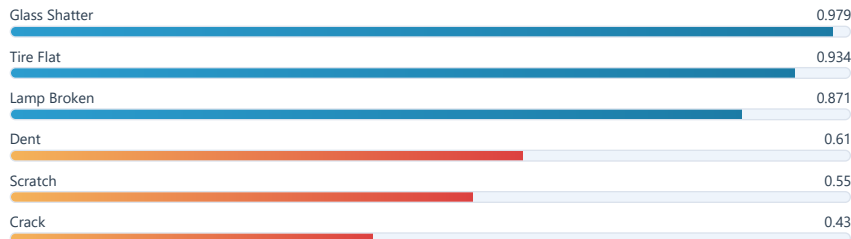
WORKING PIPELINE



RESULTS

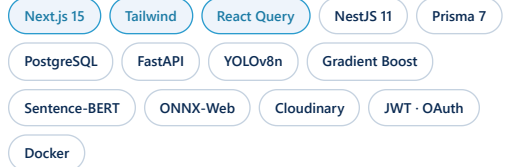
PER-CLASS MAP@0.5 (CARDD TEST SET)

TECH STACK



COST MODEL — RMSE

Gradient Boosting (selected) < Random Forest < Extra Trees < Linear Regression



KEY MODULES

12 NestJS modules · 8 Prisma models · 80+ REST endpoints
Global guards: Throttle · JWT · Roles · @RequireVerification

CONCLUSION

CARPER successfully demonstrates that a lightweight **YOLOv8n** model trained on the CarDD dataset can deliver competitive multi-class vehicle damage detection at real-time inference speed (**4.4 ms/image**), achieving an overall **mAP@0.5 of 0.732** with particularly strong performance on Glass Shatter (0.979), Tire Flat (0.934), and Lamp Broken (0.871). The integrated cost estimation module and semantic part-recommendation pipeline further extend CARPER into a practical, end-to-end automotive assessment tool. The system fulfils all three project success criteria and is presented as a fully containerised, web-deployed solution suitable for insurance automation, vehicle rental services, and private resale valuation.